

CSE 438 · Robotics Lab

University of Asia Pacific

FireFighter Robot



Presented by

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Team Members

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CSE 438 — Robotics Lab

Autonomous Firefighting Robot

Real-Time Flame Detection · Obstacle Avoidance · Autonomous Suppression

₹4,150

Budget

4WD

Chassis

3-Mode

Operation

My Contribution

Survey & Idea



Market Survey

Surveyed fire-safety sensors available in Bangladesh — selected IR Flame Sensor for cost-effectiveness (৳150) in high-heat environments.



System Definition

Defined operational logic for the 4WD chassis: three-mode hierarchy — Fire Mode → Obstacle Mode → Normal Mode.



Impact Assessment

Identified garments & manufacturing sectors as high-risk zones. Robot targets human risk reduction in fire-prone areas.

My primary role was Survey & Idea — this deck focuses on system design and rationale, not code.

Related Study & Review Material

Platform Comparison

Feature	Arduino UNO	ESP32
Cost	~₹1600	~₹2500
Real-Time I/O	✓ Excellent	✓ Good
WiFi/IoT	✗ None	✓ Built-in
PWM Channels	6	16
Community	★★★★★	★★★★☆
Our Choice	✓ Selected	—

Key References

IoT Design Pro

IR flame sensor with analog averaging achieves stable detection — adopted in our detectFlame() logic.

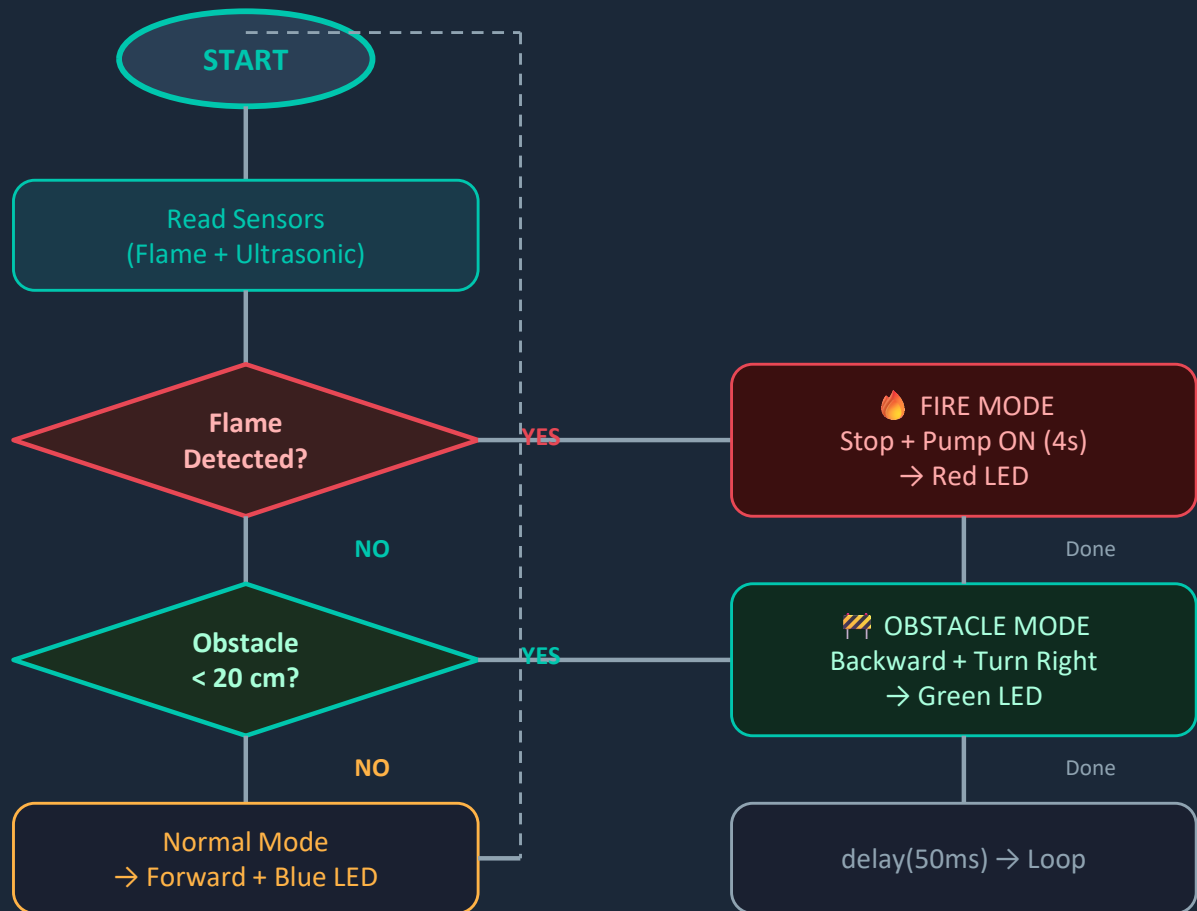
DroneBot Workshop

HC-SR04 ultrasonic sensor gives $\pm 3\text{mm}$ accuracy at 20cm threshold — validates our avoidDist setting.

Random Nerd Tutorials

L298N motor driver best-suited for 12V 4WD chassis with independent PWM speed control.

System Logic & Flow Diagram



Legend



Fire Mode (Priority 1)



Obstacle Mode (Priority 2)



Normal Mode (Default)

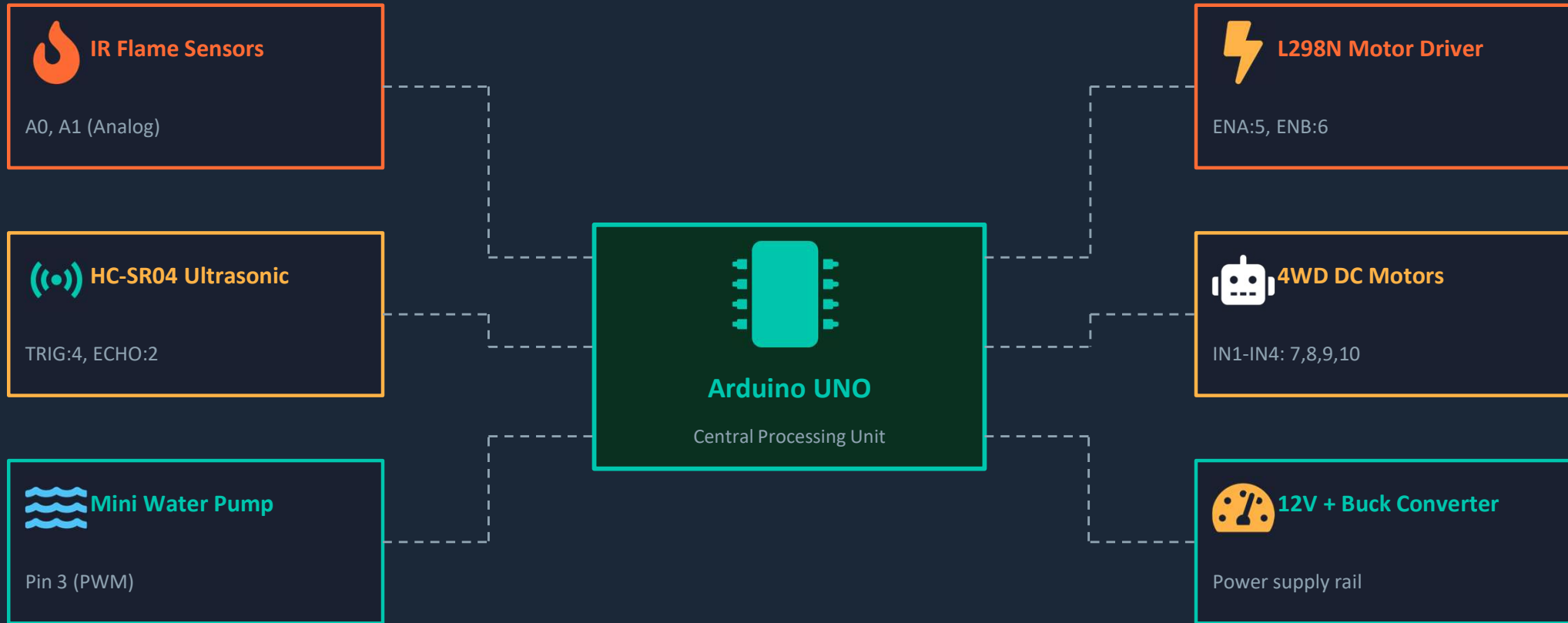


Decision Node



Process / Action

Embedded Diagram & Hardware Architecture

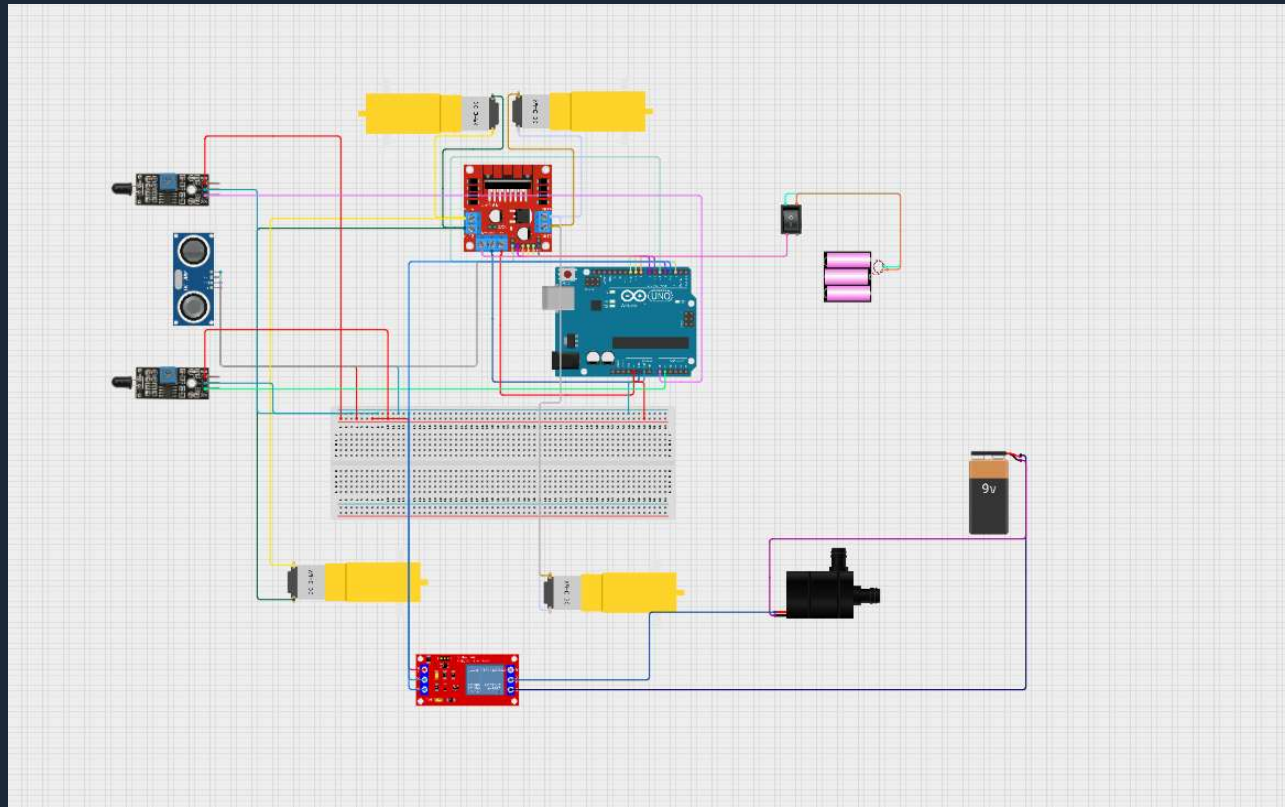


PROJECT DEMO VIDEO

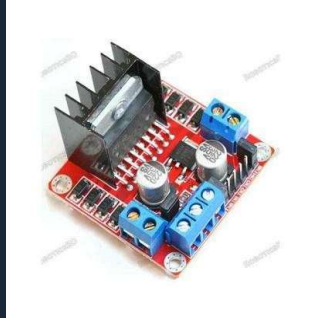


Obstacles Avoiding with firefighter bot

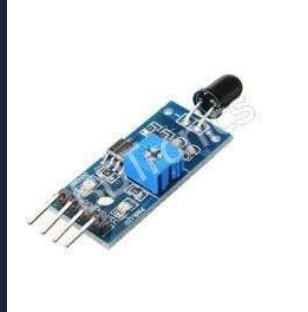
Virtual Diagram



HARDWARE IMAGES



Motor Driver



IR Flame Sensor



Ultrasonic Sensor



**Temperature &
Humidity Sensor**



Servo Motor



**Submersible Water
Pump**



**4WD Smart Robot
Chassis Kit**

HARDWARE IMAGES



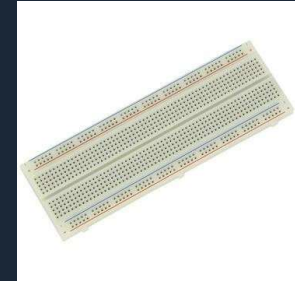
**ESP32-CAM
Module**



Li-Ion Battery



Battery Case



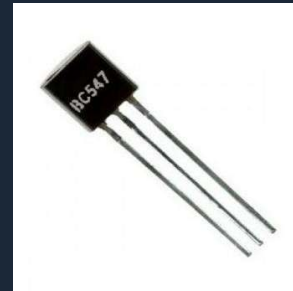
Solder-less Breadboard



**5V Single Channel Relay
Module**



Buck Converter

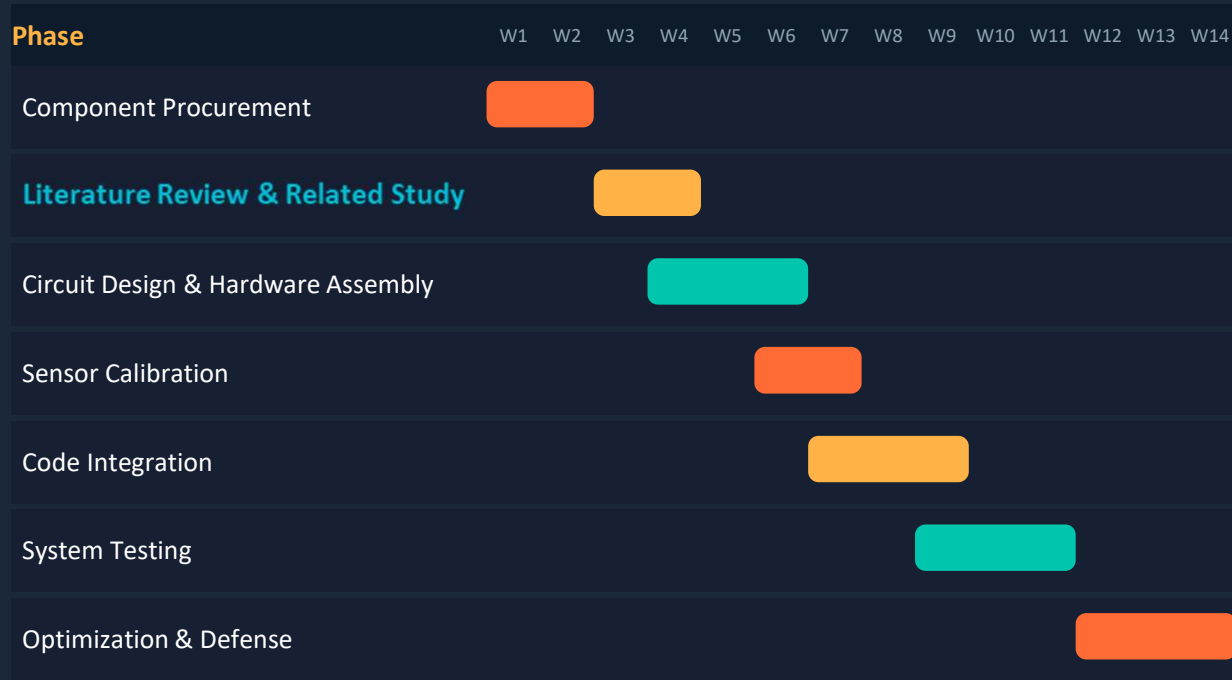


**BC547
Transistor**



**4WD Smart Robot Chassis
Kit**

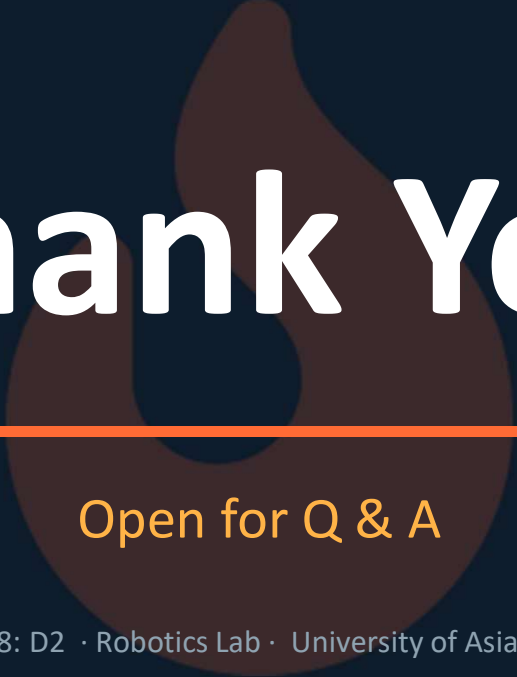
Gantt Chart & Social Impact



Social Impact

- ✓ Reduces human risk in garment & manufacturing fire incidents
- ✓ Low ₹4,150 cost enables wide deployment in SMEs
- ✓ Promotes STEM education through hands-on IoT projects
- ✓ IoT-ready design — upgradeable to remote monitoring

Thank You



Open for Q & A

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Survey & Idea · FireFighter Robot Project